EXPERIMENT 25: UNIX I/O System Calls

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <dirent.h>

int main() {

  int fd;
  char buffer[100];

  struct stat fileStat;

  DIR *dir;
  struct dirent *entry;

  // Create and open file

  fd = open("foo.txt", O_CREAT | O_RDWR | O_TRUNC, 0644);

  if (fd == -1) {
    perror("File opening failed");
    return 1;
  }

  printf("File created successfully.\n");

  // Write data
```

```
write(fd, "Operating System", 17);

printf("Data written successfully.\n");

// Demonstrate fcntl()

int flags = fcntl(fd, F_GETFL);

if (flags != -1) {
   printf("fcntl(): File status flags retrieved.\n");
}

// Demonstrate lseek()

lseek(fd, 0, SEEK_SET);

printf("lseek(): File pointer moved to beginning.\n");

// Read data

int bytes = read(fd, buffer, sizeof(buffer) - 1);

buffer[bytes] = '\0';

printf("File contents: %s\n", buffer);

// Demonstrate stat()

if (stat("foo.txt", &fileStat) == 0) {
```

```
        printf("stat(): File information retrieved.\n");

        printf("File size: %ld bytes\n",
               (long)fileStat.st_size);
    }

    close(fd);

    // Demonstrate opendir() and readdir()

    dir = opendir(".");

    if (dir == NULL) {
        perror("Directory opening failed");
        return 1;
    }

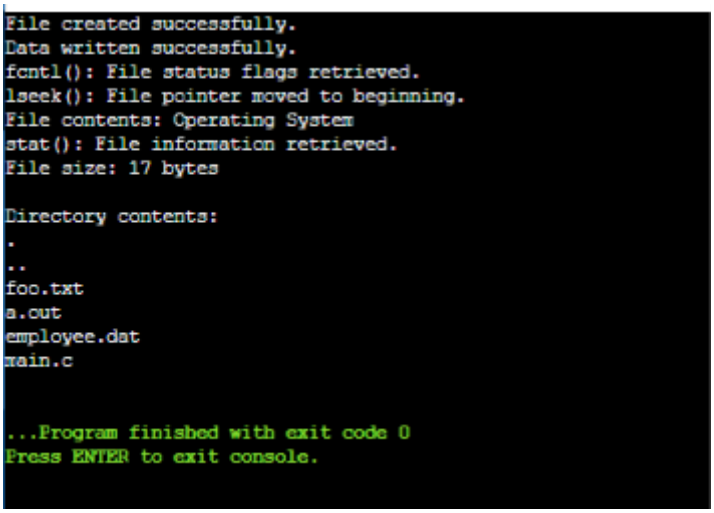
    printf("\nDirectory contents:\n");

    while ((entry = readdir(dir)) != NULL) {
        printf("%s\n", entry->d_name);
    }

    closedir(dir);

    return 0;
}
```

Output:


File created successfully.
Data written successfully.
fcntl(): File status flags retrieved.
lseek(): File pointer moved to beginning.
File contents: Operating System
stat(): File information retrieved.
File size: 17 bytes
Directory contents:
.
..
foo.txt
a.out
employee.dat
main.c
...Program finished with exit code 0
Press ENTER to exit console.